

# Rahul Rawat

Working Student, Data Science and AI for X-Ray Technology

## PERSONAL DETAILS

|               |                                                                      |
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| Address       | C2 16, 68159 Mannheim, Germany                                       |
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| Email         | rahulrawat2r@gmail.com                                               |
| LinkedIn      | linkedin.com/in/rahulrawat2r                                         |
| GitHub        | github.com/rahulrawat20022002                                        |
| Portfolio     | rah-portfolio.pages.dev                                              |
| Date of birth | 20 February 2002                                                     |
| Nationality   | Indian, student visa with valid work permit                          |
| Availability  | Werkstudent 20 hours per week immediately, full time from April 2027 |

## PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on Python machine learning, deep learning benchmarking on unstructured data, and honest evaluation of models on imbalanced healthcare datasets. I have shipped a fairness by design credit scoring system on real data with SHAP driven subgroup analysis, an end to end recursive time series pipeline at eRay GmbH benchmarking six models head to head with strict anti leakage rules, and a bachelor thesis comparing six classifiers on a clinical dataset with **10 fold** cross validation, choosing ROC AUC over accuracy for a **65 to 35** imbalanced target. Proficient in Python, scikit learn, and modern ML frameworks, I am the right fit for a Working Student position on Data Science and AI for X-Ray Technology at Siemens Healthineers in Forchheim.

## PROFESSIONAL EXPERIENCE

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| Oct 2025 to Mar 2026<br>Heidelberg | <b>eRay GmbH</b><br>Data Scientist <ul style="list-style-type: none"><li>Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.</li><li>Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric <b>80 percent</b> prediction intervals for decision support under uncertainty.</li><li>Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.</li><li>Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.</li><li>Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.</li></ul> |
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**Technologies:** Python, CatBoost, Prophet, scikit learn, MICE

## PERSONAL PROJECTS

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2025

### CreditIQ: Fairness by Design Credit Scoring

- Raised the model's Disparate Impact ratio from a failing **0.79** to a compliant **0.88**, measured against the EU AI Act and AGG **80 percent** fairness threshold, using AIF360 mitigation and threshold calibration on a real credit dataset.
- Diagnosed a hidden intersectional bias where younger women were still penalised even after single axis age bias was fixed, measured through SHAP driven subgroup analysis, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from **44 percent** to **16.7 percent** while holding accuracy at **75 percent**, measured on the held out test split, by documenting the fairness accuracy trade off as a deliberate and regulator defensible decision rather than a hidden compromise.
- Shipped a Streamlit decision support tool that gives finance managers a recommendation plus a plain language LLM generated explanation, measured against GDPR Article 22 and EU AI Act Article 14 human in the loop requirements, and backed the pipeline with unit tests at **100 percent** branch coverage and a full regulatory write up.

**Technologies:** *Python, scikit learn, AIF360, SHAP, Streamlit*

2025

### Hybrid RAG Orchestrator with Agentic Routing

- Delivered a modular Retrieval Augmented Generation system that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, measured by clean end to end responses across all three routes, using a custom decision making router on top of Llama **3.1 8b** via Groq and LangChain orchestration.
- Built a persistent semantic memory over PDF and vector data, measured by consistent retrieval on cross session queries, using ChromaDB with local persistence and HuggingFace MiniLM L6 v2 embeddings.
- Kept multi turn context intact for the LLM, measured by coherent follow up answers rather than isolated single shot responses, by adding a stateful MemoryAgent directly into the inference pipeline.
- Shipped the whole system as a Streamlit interface with end to end ownership, measured by a working deployed prototype, using Groq inference and a modular routing layer that can be extended per new intent.

**Technologies:** *Python, LangChain, Llama 3.1 8b via Groq, ChromaDB, HuggingFace MiniLM L6 v2, Streamlit*

2025

### Movie Analytics and ML Pipeline on GCP

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, measured against the ML early warning view, and secured the system with a least privilege service account and Secret Manager.

**Technologies:** *GCP BigQuery, Cloud Run, GCS, Cloud Scheduler, BigQuery ML, Python, SQL, Looker Studio*

EDUCATION

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| Apr 2025 to Present<br>Heidelberg    | <b>M.Sc. Data Science and Analytics</b><br>SRH University of Applied Sciences Heidelberg, GPA 1.9                    |
| 2019 to 2023<br>Greater Noida, India | <b>Bachelor of Technology in Computer Science</b><br>GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10 |

RESEARCH AND THESIS

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| 2022 to 2023<br>Greater Noida, India | <b>Bachelor Thesis, Diabetes Prediction Using Machine Learning</b><br>GL Bajaj Institute of Technology and Management, IEEE style paper <ul style="list-style-type: none"><li>Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of <b>768 patients</b>, using <b>10 fold</b> cross validation and per model confusion matrices to make the model comparison auditable.</li><li>Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.</li><li>Chose ROC AUC over accuracy as the headline metric for a <b>65 to 35</b> imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.</li><li>Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.</li></ul> <b>Technologies:</b> <i>Python, scikit learn, Pandas, Seaborn</i> |
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CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

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|----------------|----------------------------|
| <b>English</b> | Fluent, written and spoken |
| <b>German</b>  | B1 in progress toward B2   |
| <b>Hindi</b>   | Native                     |

Mannheim, 26 July 2026  
Rahul Rawat